

Faheem Shah Umer Vattam Kandathil

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WORK EXPERIENCE

Werkstudent

IAV GmbH

Sep 2024 – Present

Bavaria, Germany

- Developed three RAG-based AI applications for document and ECU signal analysis, used in production by an estimated 50+ users.
- Built agentic retrieval across PDFs, OneNote, email, Word, Excel, ECU data, and signal documentation, reducing reliance on manual e-handbook searches.
- Designed diagnostic logic for fuel tank leakage detection as part of functional safety development on ECU software.
- Measured and analysed ECU signals using ETAS MDA and ETAS INCA, validating ECU safety logic through simulation, numerical signal analysis, and prototype vehicle testing for an automotive OEM.

Masterarbeit

KTmfk – Lehrstuhl für Konstruktionstechnik | FAU Erlangen Nuremberg

Apr 2026 – Present

- Developed and evaluated a LangGraph-based framework orchestrating seven specialized agents for tolerance management in product development.
- Validated the prototype on several CAD files while integrating engineering knowledge retrieval, CAD/STEP and PMI processing, Monte Carlo tolerance analysis, visualization, FE meshing, and Siemens VisVSA execution.
- Implemented RAG, structured engineering artifacts, deterministic Python calculations, traceable agent handoffs, and human-in-the-loop clarification.

Projektarbeit

KTmfk – Lehrstuhl für Konstruktionstechnik | FAU Erlangen Nuremberg

Apr 2025 – Sep 2025

Grade: 1.0

- Built a RAG knowledge base from approximately 4,000 technical documents, including books, articles, standards, project papers, and theses.
- Evaluated the prototype on 20+ engineering questions, achieving approximately 95% retrieval and answer accuracy while measuring response time and user satisfaction.

Software Engineer and Content Creator

Baabte System Technologies Pvt Ltd

Jan 2023 – Apr 2023

Calicut, India

- Built working models and created technical content for Robotics, IoT, and AI projects using Arduino, MicroPython, and Raspberry Pi.
- Tools: Jira, Arduino IDE, Fritzing, MicroPython, NodeMCU, ESP32/ESP8266.

Digital Twin Benchmarking in Electronics

Caresoft Global

Apr 2022 – Jul 2022

Coimbatore, India

- Benchmarked IC components, researched architecture, manufacturer, and production cost, and compiled a final report on an electric vehicle module for use in future upgrades.

Intern

Technologies Global Pvt. Ltd

Feb 2021 – Mar 2021

Bangalore, India

- Trained in ML algorithms and Python using TensorFlow, Keras, and Pandas in Jupyter Notebook.

EDUCATION

Master's in Electromobility ACES | FAU Erlangen Nuremberg

| Present | Current GPA: 1.6

- Specializing in AI, autonomous driving, and sustainable mobility.
- Strong knowledge of electromobility technologies, including electronics, AI, and machine learning.
- Experience with simulation and modelling of electromobility systems.
- Proficient in Python.

EDUCATION

Bachelor in Mechatronics Engineering | HICET Coimbatore

| Jul 2018 – May 2022

- CGPA: 8.2 out of 10
- Proficient in Python, C++, MATLAB, MSC ADAMS, Arduino and Windows Office.
- Experience with Mechatronics systems, including sensors, actuators, and controllers.

SKILLS

AI / Agentic Systems:	LLMs, RAG, Multi-Agent Systems, LLMOps, LangChain, LangGraph, Hugging Face, Qdrant, Embeddings, Human-in-the-Loop Workflows, SpaCy, NLTK, TensorFlow, Keras.
Engineering / Simulation:	Tolerance Management, CAD/STEP and PMI Processing, Monte Carlo Tolerance Analysis, OpenCascade/XCAF, Gmsh, Siemens VisVSA, Matplotlib, MATLAB, MSC ADAMS.
Automotive / Embedded:	Functional Safety (ECU), ETAS MDA, ETAS INCA, Digital Twin, Arduino, MicroPython, Raspberry Pi, Soldering, Welding.
Programming / Tools:	Python, C, C++, PyQt5, Streamlit, Git, GitLab, Bitbucket, Jira, Confluence, Word-Press, Excel, Word.
Soft Skills:	Leadership, Teamwork, Communication, Problem-solving, Punctuality.

PROJECTS

- **SITY – Agile Workspace Presence Monitor**
Built a privacy-preserving hardware-software prototype covering approximately six rooms, with real-time occupancy displayed on a web app and physical boards. Led hardware selection, schematic design, and microcontroller programming.
Tools: GitLab, Arduino IDE, Firebase, Flask, Miro
Link: mad.tf.fau.de/research/projects/sity/
- **Portable Robotic Arm (Android-Controlled)**
Built a sub-3kg robotic arm controllable via Android. Led a team of four through the design, build, and final report.
Tools: Arduino IDE/SDK, Android Development Tools, C++, Java.
- **Scholarship Web Portal**
Built a website for students to browse and apply for scholarships in one place. Led a team of four as the sole developer and owner.
Tools: WordPress, Ajax.

SEMINAR AND WORKSHOPS

- **National Electric Kart Championship 2020**
Participated in the Virtual Round of the National Electric Kart Championship 2020 organised by Credible Future India, November 2019.
- **Humanoid Robots**
Attended a workshop on “Humanoid Robots” conducted by the School of Engineering, CUSAT, in association with IIT-Bombay, January 2020.

CERTIFICATIONS & TRAINING

- Completed *Foundation: Introduction to LangGraph* by LangChain Academy—covered agentic AI workflows and multi-agent orchestration in Python; [certificate](#).
- Completed courses on Introduction to Self-Driving Cars (University of Toronto) and Intelligent Machining (University of Buffalo and University of New York), both offered by Coursera in August 2020.
- Awarded a Certificate of Appreciation for creating a public website named “coregarage.com” by the Department of Mechatronics, Hindustan College of Engineering and Technology.

LANGUAGES

German: A2 level

English: IELTS band 7